

Problem

A parallel resonance circuit has a resistance of $2\text{ k}\Omega$ and half-power frequencies of 86 kHz and 90 kHz . Determine:

(a) the capacitance

(b) the inductance

(c) the resonant frequency

(d) the bandwidth

(e) the quality factor

Step-by-step solution

Step 1 of 6

(a)

Determine the value of bandwidth.

$$\begin{aligned} B &= \omega_2 - \omega_1 \\ &= 2\pi(f_2 - f_1) \end{aligned}$$

Substitute 90 kHz for f_2 and 86 kHz for f_1 .

$$\begin{aligned} B &= 2\pi(90\text{ kHz} - 86\text{ kHz}) \\ &= 2\pi(4\text{ kHz}) \\ &= 2\pi(4 \times 10^3) \\ &= 8\pi\text{ krad/s} \end{aligned}$$

Determine the value of resonance frequency.

$$\begin{aligned} \omega_0 &= \frac{\omega_1 + \omega_2}{2} \\ &= \frac{2\pi(f_1 + f_2)}{2} \\ &= \pi(f_1 + f_2) \end{aligned}$$

Substitute 90 kHz for f_2 and 86 kHz for f_1 .

$$\begin{aligned} \omega_0 &= \pi(90\text{ kHz} + 86\text{ kHz}) \\ &= 176\pi\text{ krad/s} \end{aligned}$$

Step 2 of 6

Write the relation between capacitance and bandwidth.

$$B = \frac{1}{RC}$$

$$C = \frac{1}{BR}$$

Substitute $8\pi \text{ krad/s}$ for B and $2 \text{ k}\Omega$ for R .

$$\begin{aligned} C &= \frac{1}{(8\pi \times 10^3)(2 \times 10^3)} \\ &= 19.89 \times 10^{-9} \text{ F} \\ &= 19.89 \text{ nF} \end{aligned}$$

Therefore, the value of capacitance is $\boxed{19.89 \text{ nF}}$.

Step 3 of 6

(b)

Write the expression for resonance frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0^2 = \frac{1}{LC}$$

$$L = \frac{1}{\omega_0^2 C}$$

Substitute $176\pi \text{ krad/s}$ for ω_0 and 19.89 nF for C .

$$\begin{aligned} L &= \frac{1}{(176\pi \text{ krad/s})^2 (19.89 \text{ nF})} \\ &= \frac{1}{(176\pi \times 10^3)^2 (19.89 \times 10^{-9})} \\ &= 164.45 \times 10^{-6} \\ &= 164.45 \mu\text{H} \end{aligned}$$

Therefore, the value of inductance is $\boxed{164.45 \mu\text{H}}$.

Step 4 of 6

(c)

Determine the value of resonance frequency.

$$\begin{aligned} \omega_0 &= \frac{\omega_1 + \omega_2}{2} \\ &= \frac{2\pi(f_1 + f_2)}{2} \\ &= \pi(f_1 + f_2) \end{aligned}$$

Substitute 90 kHz for f_2 and 86 kHz for f_1 .

$$\begin{aligned} \omega_0 &= \pi(90 \text{ kHz} + 86 \text{ kHz}) \\ &= 176\pi \text{ krad/s} \\ &= 552.9 \text{ krad/s} \end{aligned}$$

Therefore, the value of resonance frequency is $\boxed{552.9 \text{ krad/s}}$.

Step 5 of 6

(d)

Determine the value of bandwidth.

$$\begin{aligned} B &= \omega_2 - \omega_1 \\ &= 2\pi(f_2 - f_1) \end{aligned}$$

Substitute 90 kHz for f_2 and 86 kHz for f_1 .

$$\begin{aligned} B &= 2\pi(90 \text{ kHz} - 86 \text{ kHz}) \\ &= 2\pi(4 \text{ kHz}) \\ &= 2\pi(4 \times 10^3) \\ &= 25.13 \text{ krad/s} \end{aligned}$$

Therefore, the value of band width is 25.13 krad/s.

Step 6 of 6

(e)

Determine the value of quality factor.

$$Q = \frac{\omega_0}{B}$$

Substitute 176π krad/s for ω_0 and 8π krad/s for B .

$$\begin{aligned} Q &= \frac{176\pi \text{ krad/s}}{8\pi \text{ krad/s}} \\ &= 22 \end{aligned}$$

Therefore, the value of quality factor is 22.